A Stone Duality Route to the Observable Extension Theorem

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Abstract

We prove that a compatible family of finitely additive charges on a directed system of Boolean algebras extends uniquely to a σ-additive measure on the generated σ-algebra, using Stone duality as the principal mechanism. The argument proceeds in two steps. First, Stone duality converts the directed system of Boolean algebras into a cofiltered inverse system of compact Hausdorff spaces; compactness of the inverse limit then yields a σ-additive measure on the Borel algebra of the limit space without any countable additivity assumption on the input charges. Second, the Borel algebra of the inverse limit is identified with the observable σ-algebra generated by the cylinder sets, under two conditions drawn from the broader programme: discriminability of the query system and collective exhaustion of the charge family.

The result gives a second, independent proof of the observable extension theorem established in [5] by Carathéodory methods. The two routes complement each other: the Carathéodory route assumes σ-additivity of the marginals and derives the global measure directly; the Stone route assumes only finite additivity and derives σ-additivity from topological compactness. Under shared hypotheses the two measures coincide.

The proof makes explicit a role for discriminability — the condition that queries separate points of the sample space — as a structural prerequisite for the identification of the observable σ-algebra with the Borel algebra of the Stone space. Collective exhaustion, previously identified as irreducible in [5], reappears here as the condition that the Stone measure is supported on the principal ultrafilters, i.e., on the image of the sample space inside its Stone compactification.

1 Introduction

A central problem in the foundations of probability is the extension of consistent finite-dimensional distributions to a global measure on an infinite-dimensional space. The classical solution, due to Kolmogorov, requires the marginals to be σ-additive from the outset and imposes topological conditions (standard Borel spaces) to guarantee tightness. A more recent approach, developed in [5], replaces these topological hypotheses with structural conditions on a system of observable queries and derives the global measure by Carathéodory extension.

The present paper offers a third route, using Stone duality. The key insight is that a Boolean algebra B and its Stone space $\mathrm{St}(B)$ carry equivalent information: a finitely additive charge on B corresponds bijectively to a regular Borel measure on the compact Hausdorff space $\mathrm{St}(B)$. On compact spaces, regular Borel measures are automatically σ-additive. So finite additivity on a Boolean algebra, combined with Stone duality, yields σ-additivity for free on the Stone space side.

When the Boolean algebras form a directed system, Stone duality converts this into an inverse system of compact Hausdorff spaces. The Stone space of the direct limit is the inverse limit of the individual Stone spaces (Proposition 1). A compatible family of charges on the directed system therefore assembles into a σ-additive measure on the inverse limit (Proposition 2).

The remaining task is to identify the Borel algebra of the inverse limit with the observable σ-algebra $\sigma(\mathcal{C})$ generated by all cylinder sets. This identification has two parts. The structural part (Step B1, Proposition 3) uses the fact that the clopens of the Stone space pull back under the Stone embedding $\mathrm{pure} : \Omega \to \mathrm{St}(\mathcal{C})$ to exactly the cylinder sets; discriminability ensures the embedding is injective. The measure-theoretic part (Step B2, Proposition 4) uses collective exhaustion to show that no mass escapes to the non-principal ultrafilters in $\mathrm{St}(\mathcal{C})$.

Relation to prior work. The individual components of this argument are known. The correspondence between charges on a Boolean algebra and measures on its Stone space is made explicit in [1], Section 6. The extension of compatible measures on an inverse system of compact spaces is due to Choksi [2]. The Yosida–Hewitt decomposition [7] identifies the purely finitely additive component of a charge with mass on non-principal ultrafilters. What appears to be new is the assembly of these components into a single theorem for directed systems of Boolean algebras arising from query systems, with the precise roles of discriminability and collective exhaustion identified.

Organisation. Section 2 fixes notation and states the three hypotheses. Section 3 proves Step A. Section 4 constructs the inverse limit measure. Sections 5 and 6 prove the two parts of Step B. Section 7 assembles the main theorem.

2 Setup and hypotheses

2.1 Query systems

Definition 1 (Query system)**.** A *query system* is a tuple $(\iota, \leq, \Omega, \{(\mathrm{Out}(i), \mathcal{F}_i)\}_{i\in\iota}, \{\mathrm{eval}_i\}_{i\in\iota}, \{\pi_{ij}\}_{i\leq j})$ consisting of:

- a nonempty partially ordered set $(\iota, \leq)$, the *index set*;
- a set Ω, the *sample space*;
- for each $i \in \iota$, a measurable space $(\mathrm{Out}(i), \mathcal{F}_i)$, the *outcome space at level* i;
- for each $i \in \iota$, a map $\mathrm{eval}_i : \Omega \to \mathrm{Out}(i)$, the *evaluation map at level* i;
- for each $i \leq j$ in ι, a measurable surjection $\pi_{ij} : \mathrm{Out}(j) \to \mathrm{Out}(i)$, the *refinement map*,

satisfying the *coherence condition*: $\pi_{ij} \circ \mathrm{eval}_j = \mathrm{eval}_i$ for all $i \leq j$.

2.2 Cylinder sets and the observable σ-algebra

Definition 2 (Cylinder sets)**.** For $i \in \iota$ and $A \in \mathcal{F}_i$, the *level-i cylinder with base A* is

$$\mathrm{Cyl}(i, A) \;=\; \{\, \omega \in \Omega : \mathrm{eval}_i(\omega) \in A \,\}.$$

The *cylinder generating family* is

$$\mathcal{C} \;=\; \big\{\, \mathrm{Cyl}(i, A) : i \in \iota,\ A \in \mathcal{F}_i \,\big\}.$$

For each $i \in \iota$, let B_i denote the Boolean algebra of subsets of Ω generated by $\{\mathrm{Cyl}(i, A) : A \in \mathcal{F}_i\}$. Then $\mathcal{C} = \bigcup_{i\in\iota} B_i$.

The *observable σ-algebra* is $\sigma(\mathcal{C})$, the σ-algebra on Ω generated by all cylinder sets.

Remark 1. The coherence condition gives, for $i \leq j$,

$$\mathrm{Cyl}(i, A) \;=\; \mathrm{Cyl}(j, \pi_{ij}^{-1}(A)),$$

so the same subset of Ω can be described as a cylinder at any level $j \geq i$.

2.3 Connecting maps

For $i \leq j$, define the *connecting map*

$$\varphi_{ij} : B_i \to B_j, \qquad \varphi_{ij}(\mathrm{Cyl}(i, A)) \;=\; \mathrm{Cyl}(j, \pi_{ij}^{-1}(A)).$$

This is a well-defined injective Boolean algebra homomorphism (Lemma 1 below). The family $\{B_i, \varphi_{ij}\}$ is a directed system of Boolean algebras with injective connecting maps.

2.4 Compatible charges

Definition 3 (Charge, compatible family)**.** A *charge* on a Boolean algebra B is a finitely additive function $\mu : B \to [0, \infty)$ with $\mu(\emptyset) = 0$.

A family of charges $\{\mu_i\}_{i\in\iota}$ with μ_i on B_i is *compatible* if

$$\mu_j(\varphi_{ij}(E)) \;=\; \mu_i(E) \qquad \text{for all } i \le j \text{ and all } E \in B_i.$$

2.5 The three hypotheses

The main theorem requires three conditions on the query system and charge family.

Definition 4 (EvalSurjective)**.** The query system satisfies *EvalSurjective* if $\mathrm{eval}_i : \Omega \to \mathrm{Out}(i)$ is surjective for every $i \in \iota$.

Definition 5 (Discriminability)**.** The query system *discriminates points* if for every $\omega \neq \omega'$ in Ω there exists $E \in \mathcal{C}$ with $\omega \in E$ and $\omega' \notin E$.

Definition 6 (Collective Exhaustion)**.** A charge μ on $\mathcal{C}$ (the common extension of a compatible family $\{\mu_i\}$) satisfies *collective exhaustion* (CE) if: whenever $E_n \in \mathcal{C}$ and $E_1 \supseteq E_2 \supseteq \cdots$ with $\bigcap_n E_n = \emptyset$, we have $\mu(E_n) \to 0$.

Remark 2. CE is not derivable from any structural or algebraic condition on the query system alone [5]. It is an irreducible condition on the charge family: it rules out purely finitely additive charges, which satisfy every finite consistency condition while assigning persistent mass to sequences that vanish in the limit.

3 Step A: the Stone space of the direct limit

We first verify that the connecting maps are injective, then invoke the standard Stone duality fact.

Lemma 1 (Injectivity of connecting maps)**.** *If the query system satisfies* EvalSurjective*, then* $\varphi_{ij} : B_i \to B_j$ *is an injective Boolean algebra homomorphism for every* $i \le j$*.*

Proof. **Step 1: π_{ij} is surjective.** Let $x \in \mathrm{Out}(i)$. Since eval_i is surjective, there exists $\omega \in \Omega$ with $\mathrm{eval}_i(\omega) = x$. By coherence, $\pi_{ij}(\mathrm{eval}_j(\omega)) = \mathrm{eval}_i(\omega) = x$, so $x \in \mathrm{im}(\pi_{ij})$.

Step 2: φ_{ij} is injective. Suppose $\varphi_{ij}(\mathrm{Cyl}(i, A)) = \varphi_{ij}(\mathrm{Cyl}(i, A'))$, i.e., $\mathrm{Cyl}(j, \pi_{ij}^{-1}(A)) = \mathrm{Cyl}(j, \pi_{ij}^{-1}(A'))$. This gives $\pi_{ij}^{-1}(A) = \pi_{ij}^{-1}(A')$. Since π_{ij} is surjective, the preimage map π_{ij}^{-1} is injective on subsets of $\mathrm{Out}(i)$, so $A = A'$, hence $\mathrm{Cyl}(i, A) = \mathrm{Cyl}(i, A')$. Injectivity on generators extends to the Boolean algebra B_i by the universal property. □

Proposition 1 (Step A)**.** *The Stone space of the direct limit* $\mathrm{St}\big(\bigcup_i B_i\big)$ *is homeomorphic to the inverse limit* $\varprojlim_i \mathrm{St}(B_i)$*.*

Proof. By Lemma 1, the connecting maps φ_{ij} are injective Boolean algebra homomorphisms. Under Stone duality, injective Boolean algebra homomorphisms correspond to surjective continuous maps between Stone spaces. A directed system of Boolean algebras with injective connecting maps therefore dualises to a cofiltered inverse system of compact Hausdorff spaces with surjective bonding maps.

The Stone space functor St converts colimits (direct limits) in the category of Boolean algebras to limits (inverse limits) in the category of compact Hausdorff spaces [4, II.4]. Applied to $\bigcup_i B_i$ with its injective system, this gives $\mathrm{St}\big(\bigcup_i B_i\big) \cong \varprojlim_i \mathrm{St}(B_i)$. □

4 The inverse limit measure

Proposition 2 (Inverse limit measure). *Let $\{\mu_i\}$ be a compatible family of normalised charges on $\{B_i\}$. Then there exists a unique probability measure $\hat{\mu}$ on $\mathcal{B}(\varprojlim_i \mathrm{St}(B_i))$ such that the pushforward of $\hat{\mu}$ along each projection $\varprojlim_i \mathrm{St}(B_i) \to \mathrm{St}(B_j)$ agrees with the Borel measure on $\mathrm{St}(B_j)$ corresponding to μ_j.*

Proof. **Step 1: Single-algebra correspondence.** For each i, the Boolean algebra B_i is isomorphic (via Stone duality) to the clopen algebra $\mathrm{Clop}(\mathrm{St}(B_i))$ of its Stone space. A charge μ_i on B_i therefore defines a finitely additive set function on the clopens of the compact Hausdorff space $\mathrm{St}(B_i)$. Since $\mathrm{St}(B_i)$ is compact and totally disconnected, the clopens form a base for the topology. A finitely additive charge on a base of clopens of a compact space extends uniquely to a regular Borel measure; see [1], Section 6, Theorem D (or [3], §§53–54 for the content construction). Call this measure $\hat{\mu}_i$ on $\mathrm{St}(B_i)$.

Step 2: Compatibility. The compatibility condition $\mu_j(\varphi_{ij}(E)) = \mu_i(E)$ translates, under the Stone correspondence, to the condition that $\hat{\mu}_i$ is the pushforward of $\hat{\mu}_j$ along the bonding map $\mathrm{St}(B_j) \to \mathrm{St}(B_i)$. Thus the family $\{\hat{\mu}_i\}$ is a compatible family of Borel probability measures on the inverse system $\{\mathrm{St}(B_i)\}$.

Step 3: Extension to the inverse limit. Each $\mathrm{St}(B_i)$ is compact Hausdorff. The inverse limit $\varprojlim_i \mathrm{St}(B_i)$ is compact Hausdorff (being a closed subspace of a product of compact Hausdorff spaces). By [2, Theorem 1], a compatible family of regular Borel probability measures on a cofiltered inverse system of compact Hausdorff spaces with surjective bonding maps extends uniquely to a regular Borel probability measure on the inverse limit. This gives $\hat{\mu}$ on $\mathcal{B}(\varprojlim_i \mathrm{St}(B_i))$. □

5 Step B1: the structural identification

Let $\mathrm{pure} : \Omega \to \mathrm{St}(\mathcal{C})$ denote the Stone embedding, which sends $\omega \in \Omega$ to its principal ultrafilter $\mathrm{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}$. For $E \in \mathcal{C}$, write $[E] = \{u \in \mathrm{St}(\mathcal{C}) : E \in u\}$ for the corresponding basic clopen in $\mathrm{St}(\mathcal{C})$.

Proposition 3 (Step B1: structural identification). *Suppose the query system discriminates points. Then*

$$\mathrm{pure}^{-1}\big(\mathcal{B}(\mathrm{St}(\mathcal{C}))\big) \;=\; \sigma(\mathcal{C}).$$

Proof. We show the two inclusions separately.

($\supseteq$): every cylinder set is in the pullback. For any $E = \mathrm{Cyl}(i, A) \in \mathcal{C}$,

$$\mathrm{pure}^{-1}([E]) \;=\; \{\omega \in \Omega : E \in \mathrm{pure}(\omega)\} \;=\; \{\omega \in \Omega : \omega \in E\} \;=\; E.$$

Since $[E]$ is a clopen in $\mathrm{St}(\mathcal{C})$, it is Borel. Therefore every cylinder set lies in $\mathrm{pure}^{-1}(\mathcal{B}(\mathrm{St}(\mathcal{C})))$, and since this pullback is a σ-algebra, $\sigma(\mathcal{C}) \subseteq \mathrm{pure}^{-1}(\mathcal{B}(\mathrm{St}(\mathcal{C})))$.

($\subseteq$): every element of the pullback is in $\sigma(\mathcal{C})$. The Stone space $\mathrm{St}(\mathcal{C})$ is compact and totally disconnected, so its clopens form a topological basis. The clopens of $\mathrm{St}(\mathcal{C})$ are exactly the sets $[E]$ for $E \in \mathcal{C}$ [4, I.3.2]. Since the topology is generated by these clopens, the Baire σ-algebra of $\mathrm{St}(\mathcal{C})$ — the smallest σ-algebra making all continuous functions measurable — coincides with the σ-algebra generated by the clopens $\{[E] : E \in \mathcal{C}\}$.

Remark 3. On a compact Hausdorff space the Baire σ-algebra is contained in the Borel σ-algebra, with equality when the space is second-countable. We work throughout with the Baire σ-algebra on $\mathrm{St}(\mathcal{C})$; this suffices for the measure-theoretic purposes of Section 7 because the measure $\hat{\mu}$ constructed in Section 4 is a Baire measure (it is the unique extension of a charge on the clopen algebra), and Choksi's theorem [2] applies to Baire measures on compact spaces.

Now, for any Baire set $V \in \mathcal{B}_{\mathrm{Ba}}(\mathrm{St}(\mathcal{C}))$, we have $V \in \sigma(\{[E] : E \in \mathcal{C}\})$, so $\mathrm{pure}^{-1}(V) \in \sigma(\{\mathrm{pure}^{-1}([E]) : E \in \mathcal{C}\}) = \sigma(\mathcal{C})$.

Injectivity. Discriminability implies pure is injective: if $\omega \neq \omega'$, there exists $E \in \mathcal{C}$ with $\omega \in E$ and $\omega' \notin E$, so $E \in \mathrm{pure}(\omega)$ but $E \notin \mathrm{pure}(\omega')$, hence $\mathrm{pure}(\omega) \neq \mathrm{pure}(\omega')$. Injectivity ensures that the pullback does not conflate distinct events: pure^{-1} is injective on sets in the range of $[\,\cdot\,]$. □

6 Step B2: CE and the support condition

Proposition 4 (Step B2: support condition)**.** *Let μ be a charge on $\mathcal{C}$ satisfying collective exhaustion. Then the corresponding Baire measure $\hat{\mu}$ on* $\mathrm{St}(\mathcal{C})$ *is supported on* $\mathrm{pure}(\Omega)$*, the set of principal ultrafilters.*

Proof. By the Yosida–Hewitt theorem [7], every bounded charge μ on a Boolean algebra decomposes uniquely as

$$\mu \;=\; \mu_c + \mu_p,$$

where μ_c is countably additive and μ_p is purely finitely additive (i.e., $0 \leq \nu \leq \mu_p$ and ν countably additive implies $\nu = 0$).

Under Stone duality, the countably additive part μ_c corresponds to mass concentrated on the principal ultrafilters $\mathrm{pure}(\Omega) \subset \mathrm{St}(\mathcal{C})$, while the purely finitely additive part μ_p corresponds to mass on the non-principal ultrafilters — the "phantom" points of the Stone compactification. More precisely, if $\hat{\mu} = \hat{\mu}_c + \hat{\mu}_p$ is the corresponding decomposition of the Baire measure on $\mathrm{St}(\mathcal{C})$, then $\hat{\mu}_p$ is supported on $\mathrm{St}(\mathcal{C}) \setminus \mathrm{pure}(\Omega)$; see [6], Chapter 10.

Collective exhaustion is equivalent to $\mu_p = 0$: CE requires that $\mu(E_n) \to 0$ whenever $E_n \searrow \emptyset$, which is exactly the condition that the purely finitely additive part vanishes [5], Theorem (sp1_iff). Therefore $\hat{\mu}_p = 0$, and $\hat{\mu} = \hat{\mu}_c$ is supported on $\mathrm{pure}(\Omega)$. □

7 The main theorem

Theorem 1 (Stone extension theorem)**.** *Let $(\iota, \leq, \Omega, \{\mathrm{Out}(i)\}, \{\mathrm{eval}_i\}, \{\pi_{ij}\})$ be a query system satisfying* EvalSurjective *and* Discriminability*, and let $\{\mu_i\}$ be a compatible family of normalised charges on $\{B_i\}$ satisfying* Collective Exhaustion*. Then there exists a unique probability measure P on $(\Omega, \sigma(\mathcal{C}))$ such that*

$$P(\mathrm{Cyl}(i, A)) \;=\; \mu_i(\mathrm{Cyl}(i, A)) \qquad \textit{for all } i \in \iota \textit{ and } A \in \mathcal{F}_i.$$

Proof. The proof assembles the preceding propositions along the following chain:

$$\{\mu_i\} \xrightarrow{\S 4,\,\text{Step 1}} \{\hat{\mu}_i\} \xrightarrow{\S 4,\,\text{Step 3}} \hat{\mu} \text{ on } \varprojlim_i \mathrm{St}(B_i) \xrightarrow{\text{B1}} P \text{ on } \sigma(\mathcal{C}).$$

Existence. By Proposition 1, $\mathrm{St}(\mathcal{C}) \cong \varprojlim_i \mathrm{St}(B_i)$. By Proposition 2, the compatible family $\{\mu_i\}$ extends to a Baire probability measure $\hat{\mu}$ on $\varprojlim_i \mathrm{St}(B_i) \cong \mathrm{St}(\mathcal{C})$.

By Proposition 4, $\hat{\mu}$ is supported on $\mathrm{pure}(\Omega)$. Since $\mathrm{pure} : \Omega \to \mathrm{St}(\mathcal{C})$ is injective (Discriminability), the pushforward $P = (\mathrm{pure})_* \hat{\mu}$ — the image measure under pure^{-1} on the support — is a well-defined probability measure on Ω.

By Proposition 3, pure^{-1} maps Baire sets in $\mathrm{St}(\mathcal{C})$ to sets in $\sigma(\mathcal{C})$, so P is defined on $\sigma(\mathcal{C})$. For any cylinder set $E = \mathrm{Cyl}(i, A)$,

$$P(E) \;=\; \hat{\mu}([E]) \;=\; \mu_i(\mathrm{Cyl}(i, A)),$$

where the last equality holds by the single-algebra Stone correspondence (Proposition 2, Step 1).

Uniqueness. Two probability measures on $(\Omega, \sigma(\mathcal{C}))$ that agree on all cylinder sets are equal, by the π-λ theorem applied to the π-system $\mathcal{C}$ (under the upper-directedness condition on ι, which is implied by EvalSurjective in the query system setting); see [5], Theorem (observational_determination). $\square$

Remark 4 (Relation to the Carathéodory route)**.** Under the additional assumption that each μ_i is already σ-additive (a countably additive probability measure), the measure P produced by Theorem 1 coincides with the measure produced by the Carathéodory extension in [5]. Both measures agree on $\mathcal{C}$; uniqueness (observational determination) then forces them to be equal. The Stone route thus provides an independent proof of the observable extension theorem, requiring only finite additivity of the input charges.

Remark 5 (CE is not bypassed)**.** It may appear that the compactness of the Stone space eliminates the need for Collective Exhaustion. Compactness does ensure that the measure $\hat{\mu}$ on $\mathrm{St}(\mathcal{C})$ is σ-additive. But $\hat{\mu}$ lives on the Stone space, not on Ω. The descent from $\mathrm{St}(\mathcal{C})$ to Ω requires that no mass escapes to the non-principal ultrafilters — and that is precisely CE (Proposition 4). In the Stone picture, CE is not an additional hypothesis but a support condition: it says the measure concentrates on the image of the sample space inside its Stone compactification.

References

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